HASSAN HUSSEIN SALEH

AI Engineering Student | LLMs & RAG | Computer Vision | Machine Learning Systems

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PROFESSIONAL SUMMARY

AI Engineering student with hands-on experience building secure LLM/RAG backends, medical retrieval systems, real-time computer vision applications, machine learning pipelines, and cloud-based microservices. Experienced with LangGraph, FastAPI, Qdrant, Ollama, PyTorch, OpenCV, Docker, PostgreSQL, Redis, and AWS. Built systems spanning permission-aware Text-to-SQL, grounded document chat, gesture-driven vision interfaces, predictive modeling, and graph optimization.

TECHNICAL SKILLS

Programming: Python, C++, Java, C
AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Feature Engineering, Model Evaluation, Monte Carlo Simulation
LLMs / RAG / Agents: LangGraph, Retrieval-Augmented Generation (RAG), Ollama, Qdrant, Embeddings, Reranking, Grounded Generation, Citations
Computer Vision: OpenCV, MediaPipe, Real-Time Hand Tracking, Gesture-Based Interaction
Backend / Data / Cloud: FastAPI, Flask, REST APIs, PostgreSQL, pgvector, SQLAlchemy, Alembic, SQLGlot, Redis, Celery, MinIO, Docker, AWS, Terraform
Monitoring / Tools: Prometheus, Grafana, OpenTelemetry, Streamlit, Git, Matplotlib, NetworkX, Docling

EDUCATION

B.Sc. Computer Engineering - AI Engineering | Expected June 2027
Alamein International University (AIU), Egypt

PROJECT PORTFOLIO

Multi-Tenant Text-to-SQL & Document RAG Platform | GitHub

Python | FastAPI | LangGraph | Ollama | Qdrant | PostgreSQL/pgvector | Redis | Celery | MinIO | SQLGlot | Docker Compose

- Built a secure multi-tenant backend for permission-aware Text-to-SQL, document RAG, and hybrid chat across structured and unstructured data.
- Implemented schema-aware SQL generation and validation, role/table/column/row controls, vector retrieval with citations, background processing, auditability, and observability with Prometheus/Grafana/OpenTelemetry.

MedGround AI - Full-Stack Medical RAG | GitHub

Python | FastAPI | Streamlit | Docling | Qdrant | Ollama | Qwen Embeddings/LLMs | SQLite | RAG Evaluation

- Developed a local medical RAG application with PDF ingestion, Docling parsing, hybrid chunking, embeddings, Qdrant retrieval, grounded generation, and source evidence panels.
- Added retrieval score gates, medical safety guardrails, bilingual Arabic/English querying, persistent chat/document management, and Precision@3/Precision@5 retrieval evaluation.

Air Canvas AI - Real-Time Gesture Drawing | GitHub

Python | OpenCV | MediaPipe | NumPy | Real-Time Computer Vision

- Built a real-time two-hand air-drawing application that tracks webcam hand landmarks and converts finger gestures into drawing actions.
- Implemented independent left/right-hand control, freehand and geometric shape modes, handwriting mode, brush/eraser controls, and clean image export.

FlyRank ML Internship - Applied Search Intelligence | *Ongoing internship workspace* | GitHub

Python | Pandas | Applied Machine Learning | Search/Content Analytics | Decision-Support Modeling

- Framed an explainable content-opportunity scoring problem on 30,000 pseudonymized content pages from 32 clients to prioritize review actions such as refresh, expansion, protection, pruning, or monitoring.
- Performed initial signal analysis and identified 16,262 declining pages, including 9,961 declining pages with at least 500 impressions, while explicitly separating observed patterns from causal claims.

AI-Powered Cloud Learning Platform | GitHub

Python | Flask | Docker | Redis | JWT/RBAC | AWS | Terraform | SpeechRecognition | gTTS | PyPDF2

- Architected a five-service learning platform for chat, document processing, quiz generation, speech-to-text, and text-to-speech using a microservices approach.
- Implemented JWT/RBAC, PDF/DOCX ingestion, containerized services, and reproducible AWS infrastructure provisioning with Terraform.

AI World Cup 2026 Prediction System | GitHub

Python | Pandas | NumPy | Scikit-learn | Streamlit | Matplotlib | Elo/FIFA Features | Monte Carlo Simulation

- Built a machine learning system for international football match prediction using historical results, team form, goal statistics, FIFA rankings, and Elo ratings.
- Engineered predictive features and tournament simulation logic to estimate team winning probabilities across repeated World Cup 2026 scenarios in an interactive Streamlit app.

Smart City Transportation Network Optimization - Cairo | GitHub

Python | NetworkX | Graph Algorithms | Dynamic Programming | Greedy Algorithms | Tkinter | Matplotlib

- Modeled Greater Cairo transportation as a weighted graph and implemented Dijkstra, A*, time-dependent routing, MST algorithms, and public-transit scheduling approaches.
- Built simulation/visualization modules for traffic-signal optimization, emergency-vehicle prioritization, route analysis, and smart-city transportation decision support.

CERTIFICATIONS

AWS Cloud Foundations | IBM SkillsBuild - Artificial Intelligence | Huawei HCIA-AI | ITI: OOP (C++), Core Java, JDBC, JavaFX, Java IO & Networking

LANGUAGES

Arabic: Native | English: Upper-Intermediate